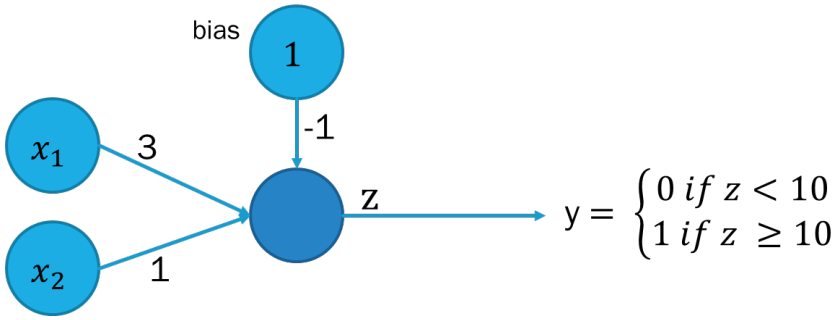
6COSC020W Applied AI

Tutorial Week 8: Artificial Neural Networks

Aim:

- Review the perceptron and forward propagation.
- Experiment with TensorFlow and neural networks.
- Train and evaluate a neural network (model).
- Understand the performance of the network.
- Improve the performance of the network by changing some hyper parameters.

1.- Using forward propagation, calculate the output (y) of the following perceptron with the given inputs:



a) x_1 = 5 and x_2= 10

(5*3) + (10*1) + (1*-1) = 15 + 10 -1 = 24; 24 >= 10 → y = 1

b) x_1 = 1 and x_2= 2

(1*3) + (2 *1) + (1*-1) = 3 + 2 – 1 = 4; 4<10 → y = 0

Compare your results with the person sitting next to you.

2.- Tutorial_Week8.ipynb.

Solutions for the Python Notebook are provided in Tutorial_Week8_solutions.ipynb